

VONEGAN, Russian Federation

WMO: 238230

Lat: 61.9561N

Lon: 64.9378E

Elev: 365

StdP: 14.50

Time Zone: 5.00 (E05)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.2	-31.8	-43.0	0.5	-37.8	-36.7	0.7	-31.2	17.8	13.4	16.1	7.4	2.8	300	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	83.0	65.7	79.6	64.2	76.1	62.7	68.1	78.9	66.1	76.6	64.2	73.6	6.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	90.1	72.9	62.0	84.2	70.6	60.3	79.1	69.3	32.6	78.3	31.0	76.6	29.6	73.3	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.2	12.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.3	-4.5	5.3	19.2	31.2	44.2	59.8	62.3	55.6	44.7	29.5	12.1	2.7
	DBStd	25.14	15.80	14.46	12.30	9.72	8.39	7.88	7.20	6.73	9.71	14.30	16.14	
	HDD50	8177	1689	1253	953	564	242	15	6	28	187	638	1136	1466
	HDD65	12780	2154	1673	1418	1013	649	198	147	300	610	1101	1586	1931
	CDD50	993	0	0	0	1	62	311	388	203	27	1	0	0
	CDD65	122	0	0	0	0	4	43	65	10	0	0	0	0
	CDH74	1190	0	0	0	0	69	436	588	97	0	0	0	0
	CDH80	257	0	0	0	0	9	82	149	17	0	0	0	0
	CDH80	257	0	0	0	0	9	82	149	17	0	0	0	0
(14) Wind	WSAvg	5.5	4.8	5.1	6.1	6.5	6.1	5.6	4.9	4.6	5.5	6.2	5.3	5.0
(15) Precipitation	PrecAvg	26.0	1.6	1.4	1.7	1.7	2.0	2.8	2.8	2.9	2.4	2.5	2.3	1.8
	PrecMax	29.8	2.6	2.5	3.5	3.2	3.8	5.1	4.7	4.8	3.9	4.4	3.5	3.0
	PrecMin	20.4	0.6	0.3	0.2	0.0	0.9	1.5	1.0	1.6	1.0	1.0	0.4	1.1
	PrecStd	2.2	0.6	0.5	0.8	0.7	0.7	1.0	1.1	0.8	0.9	0.9	0.7	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	34.1	47.0	58.9	81.3	87.0	88.1	82.4	67.7	58.3	39.8	32.3
		MCWB	28.1	31.4	36.0	48.7	61.0	68.1	68.6	66.8	55.5	50.6	38.2	30.6
	2%	DB	25.8	31.3	41.6	52.6	75.3	82.3	83.9	76.2	63.1	49.9	34.0	29.9
		MCWB	24.7	29.0	34.1	42.8	57.5	65.5	66.1	63.9	53.2	44.4	32.1	28.5
	5%	DB	21.2	28.1	38.5	49.1	69.1	79.0	80.5	72.0	59.2	46.2	31.9	26.2
		MCWB	20.0	26.1	32.4	40.6	54.5	63.2	65.5	60.4	51.6	42.2	30.2	25.0
	10%	DB	16.9	24.5	35.4	45.7	63.2	75.5	76.6	68.5	55.7	42.3	29.6	22.5
		MCWB	15.8	22.8	30.8	38.7	51.4	61.6	64.2	59.0	49.9	39.7	28.2	21.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.3	32.3	36.9	50.3	62.8	70.5	70.2	68.0	57.0	52.1	38.2	30.8
		MCDB	29.7	33.9	44.4	59.0	78.9	80.7	83.0	79.4	64.0	57.8	40.0	31.9
	2%	WB	24.8	29.2	34.8	44.6	58.8	67.7	68.5	64.8	54.5	45.5	32.6	28.7
		MCDB	25.8	31.1	40.6	50.8	72.9	78.2	79.9	73.8	61.0	49.2	33.7	30.0
	5%	WB	20.3	26.1	33.0	41.4	55.5	65.2	66.8	62.6	52.2	42.3	30.3	25.1
		MCDB	21.3	28.0	37.6	48.2	66.6	76.8	77.1	69.3	58.6	45.6	31.8	26.0
	10%	WB	16.0	22.7	31.2	38.7	52.3	63.1	65.2	60.4	50.1	39.8	28.2	21.6
		MCDB	17.0	24.4	35.3	44.6	62.8	73.2	74.7	66.8	54.2	42.3	29.7	22.5
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	15.2	17.5	16.6	18.6	19.8	17.3	16.7	13.6	10.1	12.2	12.5
		MCDBR	15.9	14.6	16.8	20.2	27.1	24.4	24.2	23.2	19.0	13.5	10.6	10.7
	5% WB	MCWBR	15.5	13.1	11.4	12.7	13.9	11.1	10.1	11.7	11.1	9.3	10.2	10.1
		MCWBR	16.5	14.2	15.5	17.7	24.8	21.9	21.8	19.6	17.0	12.4	10.9	10.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.242	0.292	0.337	0.371	0.375	0.389	0.419	0.399	0.350	0.313	0.256	0.230
		taud	1.973	2.041	2.017	2.090	2.334	2.367	2.276	2.310	2.417	2.334	2.055	1.915
	Ebn at Noon	Ebn at Noon	132	203	236	255	267	265	252	246	239	202	126	74
		Edh at Noon	12	24	36	41	35	35	37	33	24	18	11	7
(44) All-Sky Solar Radiation	RadAvg	RadAvg	103	373	857	1318	1558	1681	1657	1168	693	321	109	47
	RadStd	RadStd	6	29	52	167	131	118	148	111	93	40	16	4

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page